

Homework 2: Monte Carlo Estimation

1. The program generates n random numbers uniformly distributed in the $[0.0, 1.0)$ interval and calculates the proportion that those numbers that fall within $[0.0, 0.5)$. In addition, the answer will be output as a percentage.
2. If the user enters 10000, because the random numbers are always average. So the output should be 50%.
3. The first answer will change that the program calculates the proportion that those numbers that fall within $[0.0, 0.5]$. Therefore, 0.5 will be taken in account. However, it is hard to come out 0.5, so the output will be slightly larger or even the same.
- 4.

```
public static void main(String[] args) {  
    SimpleReader input = new SimpleReader1L();  
    SimpleWriter output = new SimpleWriter1L();  
  
    output.print("Number of points: ");  
    int n = input.nextInt();  
  
    int ptsInInterval = 0, ptsInSubinterval = 0;  
  
    Random rnd = new Random1L();  
  
    while (ptsInInterval < n) {  
        double x = rnd.nextDouble();  
        ptsInInterval++;  
        if (x < 0.5) {  
            ptsInSubinterval++;  
        }  
    }  
  
    double estimate = (100.0 * ptsInSubinterval) / ptsInInterval;  
    output.println("Estimate of percentage: " + estimate + "%");  
  
    input.close();  
    output.close();  
}
```